

# **WALMART WEEKLY STORE SALES FORECASTING: A DATA-DRIVEN APPROACH TO INVENTORY AND REVENUE PLANNING ACROSS A 45-STORE RETAIL NETWORK**

**Project Report Submitted in fulfillment of the  
Requirements for the Award of the Internship of  
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Data Science and AI

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*Through the platform:*



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**India Data Research Academy**

## DECLARATION

I, NITHYA PRAKASH S (Institute Roll No. 24CSR214, Enrollment No. IDRA-2026-858984), hereby declare that the capstone project report titled “WALMART WEEKLY STORE SALES FORECASTING: A DATA-DRIVEN APPROACH TO INVENTORY AND REVENUE PLANNING ACROSS A 45-STORE RETAIL NETWORK”, submitted in fulfillment of the requirements of the Data Science & AI Training Program conducted by the India Data Research Academy (IDRA), is my own original work.

The data acquisition, data-quality auditing, feature engineering, statistical testing, machine learning modelling, hyperparameter tuning, and the accompanying Jupyter notebook were carried out by me under the guidance of my supervisor. Wherever external sources, datasets, libraries, or documentation have been used, they have been duly acknowledged in the References section. This report has not been submitted, in whole or in part, for the award of any other degree, diploma, or certification.

Place: Kongu Engineering College

Date: 20 September 2026

**Signature: NITHYA PRAKASH**

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## ABSTRACT

This project applies a complete, leakage-aware data science workflow to the problem of weekly store-level sales forecasting for a 45-store Walmart retail network. The dataset comprises 6,435 weekly records spanning 5 February 2010 to 26 October 2012, with each of the 45 stores contributing exactly 143 consecutive weekly observations described by weekly sales, a holiday indicator, and four macroeconomic or environmental covariates (temperature, fuel price, consumer price index, and unemployment rate). A structural audit confirmed that the panel contains no duplicate rows, no missing values, and no irregular gaps: every store timeline advances in strict seven-day increments, so no interpolation or imputation was required. Anomaly profiling was performed using store-specific interquartile-range bounds rather than a single global threshold, which identified 300 unusual weeks (4.66% of all records); because 28.3% of these coincide with flagged holiday weeks and the remainder are physically plausible sales values, all rows were retained as genuine demand signal. Three families of forecasting-safe features were then engineered: cyclical calendar encodings, lagged sales at one-week and fifty-two-week horizons, and a four-week trailing rolling mean computed only from prior weeks. Exploratory and statistical analysis established a statistically significant holiday-week sales lift (Welch’s  $t = 2.680$ ,  $p = 0.0076$ , mean uplift 7.84%, median uplift 6.52%), a weak raw association between sales and the macroeconomic covariates ( $|r| \leq 0.106$ ), and a threefold spread in store-level volatility (coefficient of variation from approximately 0.04 to 0.23). A strictly chronological out-of-time split reserved the final 26 weeks of every store’s own history as a held-out test set, yielding 2,925 training and 1,170 test records with

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no date overlap. A Random Forest regressor tuned by randomised search with three-fold cross-validation achieved a test-period MAE of \$39,715.89, RMSE of \$62,910.81, MAPE of 3.89%, and  $R^2$  of 0.9862, improving on a last-week naive baseline by 21.92% and on a last-year seasonal-naive baseline by 26.49% on MAE. Feature-importance analysis shows that annual seasonality dominates: the 52-week lag alone accounts for 72.60% of predictive power and the two strongest lag/rolling features together account for 97.00%. The report concludes with inventory-planning recommendations anchored on year-over-year tracking, volatility-tiered safety stock, and a deployment roadmap for operationalising the model.

**Keywords:** exploratory data analysis (EDA); demand forecasting; feature engineering; out-of-time validation; random forest; hyperparameter tuning; inventory planning.

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# 1. INTRODUCTION

## 1.1 Background

Large retail chains operate on thin margins and high volumes, which makes the accuracy of their sales forecasts an operational rather than an academic concern. Every store must decide, week by week, how much stock to hold, how many staff to roster, and how much working capital to commit. An over-forecast leaves shelves and stockrooms full of merchandise that eventually has to be marked down; an under-forecast produces empty shelves, lost sales, and a customer who buys elsewhere. Because replenishment decisions must be taken before demand is observed, the forecast is the single input on which the rest of the supply chain depends. This project analyses a real, publicly circulated Walmart weekly sales panel in order to understand what actually drives store-level revenue, to quantify the contribution of holidays and macroeconomic conditions, and to build and honestly benchmark a forecasting model that respects the chronological structure of the data.

## 1.2 Context of the Problem

The dataset used in this project tracks weekly sales for 45 Walmart stores across the full period from February 2010 to October 2012. For every store-week it records the total sales achieved, whether that week contained a major United States holiday, and four contextual variables describing the economic and environmental conditions in the store's region: average temperature, regional fuel price, the consumer price index, and the regional unemployment rate. In a real planning environment, exactly this combination of internal sales history and external economic context is what a demand-planning team would use to set replenishment quantities, plan promotional inventory, and decide which stores require the closest supervision during peak trading weeks such as Thanksgiving and Christmas.

## 1.3 Scope of the Study

This study covers dataset profiling, structural and data-quality auditing, store-specific anomaly analysis, leakage-aware feature engineering, exploratory and inferential statistical analysis, and the development, tuning, and evaluation of a regression model for weekly sales using a strict out-of-time validation design. It does not cover department-level or SKU-level forecasting, multi-step-ahead forecasting beyond the one-week-ahead formulation, price or markdown optimisation, or a production deployment of the pipeline; these are identified as directions for future work in Section 17.3. Figure 1 summarises the end-to-end workflow that the remainder of this report follows.

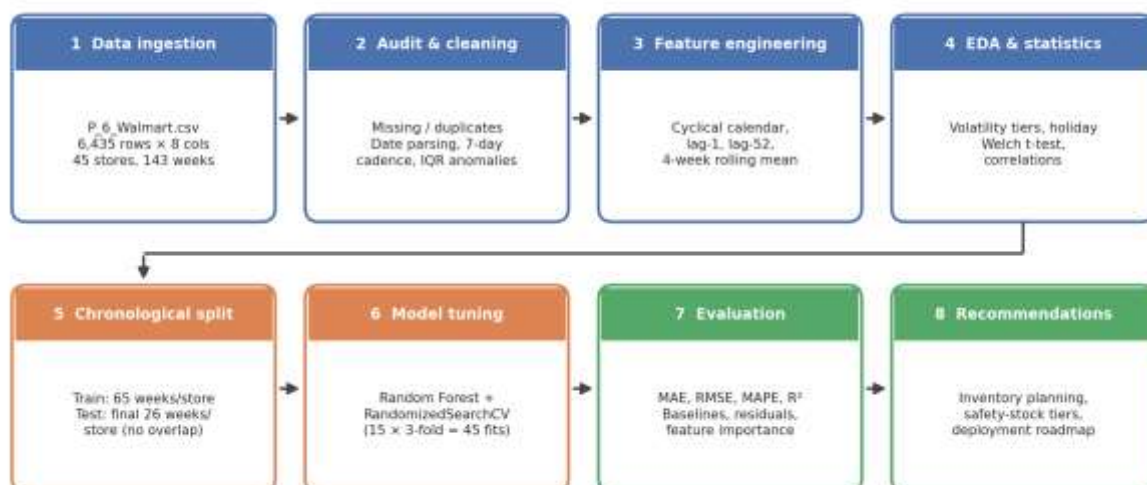


Figure 1. End-to-End Project Workflow

## 2. PROBLEM STATEMENT AND MOTIVATION

How accurately can the weekly sales of an individual Walmart store be predicted one week in advance using only information that would genuinely be available at the time the forecast is made — that is, the store's own historical sales, the calendar position of the week, the holiday indicator, and the prevailing macroeconomic conditions — and how much value does a tuned machine learning model add over the simple heuristics that planners already use in practice?

This question matters for three reasons. First, it quantifies how much of the week-to-week variation in store revenue is genuinely explainable from observable signals, which sets a realistic expectation for what any forecasting system can achieve. Second, it identifies which signals carry the information, which directly determines what a planning team should monitor and what data would be worth acquiring next. Third, and most importantly for credibility, it establishes an honest benchmark: a model that is only evaluated against itself proves nothing, so this project measures the tuned model against two naive baselines that a planner could implement with a spreadsheet, on a test period the model has never seen.

### 3. OBJECTIVES

The project pursues the following measurable objectives:

1. To profile the structure, scale, quality, and temporal continuity of the Walmart weekly sales panel before any modelling is attempted.
2. To audit the dataset for duplicates, missing values, type inconsistencies, and anomalous sales weeks, using a store-specific rather than a global outlier criterion.
3. To engineer a forecasting-safe feature set of cyclical calendar encodings, lagged sales terms, and trailing rolling statistics that never expose a week's own sales to the model predicting it.
4. To quantify the effect of holiday weeks on sales and to test whether that effect is statistically significant rather than an artefact of sampling noise.
5. To measure the association between weekly sales and the macroeconomic covariates (consumer price index, unemployment, fuel price) and the environmental covariate (temperature).
6. To build, tune, and evaluate a Random Forest regression model for weekly sales using a strictly chronological out-of-time train/test split.
7. To benchmark the tuned model against a last-week naive baseline and a last-year seasonal-naive baseline on an identical held-out period.
8. To translate the analytical and modelling results into practical inventory, safety-stock, and deployment recommendations.

## 4. RESEARCH QUESTIONS

1. What are the major structural characteristics, data-quality properties, and temporal completeness of the Walmart weekly sales panel?
2. Do holiday weeks generate a statistically significant sales lift relative to non-holiday weeks, and how large is that lift in both absolute and standardised terms?
3. How strongly are the macroeconomic and environmental covariates associated with weekly sales at the pooled panel level, and what does that imply for their usefulness as predictors?
4. How much does sales volatility differ across the 45 stores, and what does that difference imply for inventory policy?
5. How accurately can a tuned Random Forest model predict weekly sales on a genuinely unseen future period, and by how much does it improve on naive baselines?
6. Which engineered features drive the model's predictions, and does the resulting importance ranking agree with the patterns found in the exploratory analysis?

## 5. DATASET DESCRIPTION AND DATA UNDERSTANDING

### 5.1 Data Source

The dataset was provided through the IDRA Learning Management System for the capstone project on Walmart store sales forecasting, as the file P\_6\_Walmart.csv. It is a store-level weekly panel covering the complete trading history of 45 stores over approximately two years and nine months. No external data was merged into the analysis; this report and the accompanying notebook rely solely on the assigned file.

### 5.2 Dataset Size and Structure

The raw dataset contains 6,435 rows and 8 columns. Table 1 summarises its structure.

**Table 1. Summary of the Dataset**

| Attribute              | Value                                          |
|------------------------|------------------------------------------------|
| Source                 | LMS-provided dataset, P_6_Walmart.csv          |
| Rows                   | 6,435                                          |
| Columns                | 8                                              |
| Time period            | 5 February 2010 – 26 October 2012              |
| Observations per store | 143 consecutive weekly records                 |
| Unique stores          | 45                                             |
| Holiday weeks          | 450 records (6.99% of the panel; 10 per store) |
| Non-holiday weeks      | 5,985 records (93.01% of the panel)            |
| Target variable        | Weekly_Sales (continuous, USD per store-week)  |
| Granularity            | One row per Store–Week combination             |
| Frequency              | Weekly, strictly seven-day spacing             |

Weekly\_Sales, the total revenue achieved by a given store in a given week, was selected as the target variable because it is the quantity that drives replenishment, staffing, and working-capital decisions. The panel is perfectly balanced: all 45 stores contribute exactly 143 weekly observations, so no store is over- or under-represented in either the training or the test partition.

## 5.3 Data Dictionary

Table 2 documents the role and type of each raw column used in the analysis.

**Table 2. Data Dictionary**

| Column       | Type                     | Role                                                |
|--------------|--------------------------|-----------------------------------------------------|
| Store        | Integer (1–45)           | Store identifier — feature                          |
| Date         | Date (string → datetime) | Week-ending date of the record                      |
| Weekly_Sales | Float                    | Total sales that week (USD) — TARGET                |
| Holiday_Flag | Binary (0/1)             | 1 if the week contains a major US holiday — feature |
| Temperature  | Float                    | Average regional temperature (°F) — feature         |
| Fuel_Price   | Float                    | Regional fuel price (USD per gallon) — feature      |
| CPI          | Float                    | Consumer Price Index — feature                      |
| Unemployment | Float                    | Regional unemployment rate (%) — feature            |

## 5.4 Panel Structure and Temporal Coverage

Because the dataset is a panel rather than an independent, identically distributed table, its temporal structure determines what forms of validation are legitimate. Each of the 45 stores carries its own uninterrupted weekly time series, and the series all begin and end on the same dates. This has two consequences that shape the rest of the analysis. First, a 52-week lag is defined for every store from week 53 onwards, which makes year-over-year seasonality directly available as a feature. Second, any validation split must be made along the time axis rather than at random, because a random split would place future weeks in the training set and allow the model to learn from information that would not exist at forecast time. The split design adopted in response is described in Section 7.3.

## 6. DATA CLEANING

Before any feature engineering, the dataset was audited column by column for missing values, duplicate rows, incorrect data types, irregular time gaps, and anomalous sales weeks. The panel proved to be structurally sound at source, which is documented explicitly below rather than assumed.

### 6.1 Missing Values

Calling `df.isna().sum()` across all eight columns returned zero for every column, and `df.info()` confirmed 6,435 non-null entries in each. No imputation was therefore necessary in the raw data. Missing values were introduced only later, deliberately, during feature engineering (Section 10), as an unavoidable consequence of computing lagged and rolling statistics that are undefined at the start of each store's series; those gaps are handled by row removal rather than imputation, for the reasons explained in Section 10.

### 6.2 Duplicate Records

The call `df.duplicated().sum()` returned 0. No duplicate store-week rows were present, so no de-duplication step was required. This is an important check for a panel dataset, because a duplicated store-week would silently corrupt both the lag features and the chronological split.

### 6.3 Data Types and Panel Continuity

The Date column was loaded as a string and converted to a true `datetime64` type using `pd.to_datetime()` with an explicit day-first format, after which the frame was sorted by Store and then Date so that each store's series is in strict chronological order. Panel continuity was then verified rather than assumed: the set of unique date differences within every store timeline was computed, and the result contained exactly one value, a gap of seven days. Every store therefore has an unbroken weekly cadence with no missing or duplicated weeks, so no reindexing, interpolation, or calendar reconstruction was needed.

### 6.4 Anomaly Profiling

Retail sales are naturally spiky around Thanksgiving, Christmas, the Super Bowl, and Labor Day, and the 45 stores operate at very different absolute sales levels — store means in this panel range from roughly \$0.37 million to \$1.39 million per week. A single global outlier threshold would therefore flag every high-volume store's ordinary trading as anomalous while missing genuine anomalies at smaller stores. Interquartile-range bounds were instead computed separately for each store, flagging a week only when its sales fall outside that store's own lower and upper fences. Table 3 reports the outcome.

Table 3. Store-Specific IQR Anomaly Profiling

| Metric                       | Value                   | Interpretation                                  |
|------------------------------|-------------------------|-------------------------------------------------|
| Flagged weeks                | 300 (4.66% of 6,435)    | Outside the store's own $1.5 \times$ IQR fences |
| Flagged and holiday          | 85 (28.33% of flagged)  | Legitimate, recurring holiday surges            |
| Flagged, non-holiday         | 215 (71.67% of flagged) | Plausible values, no data-entry errors          |
| Flag rate, holiday weeks     | 18.89% of 450           | Holiday weeks are far more likely to be extreme |
| Flag rate, non-holiday weeks | 3.59% of 5,985          | Baseline rate for ordinary weeks                |
| Relative risk                | $5.26 \times$           | Holiday weeks flagged 5.26 times as often       |
| Rows removed                 | 0                       | All records retained as genuine signal          |

The treatment decision follows directly from this profile. Holiday weeks are flagged as extreme more than five times as often as ordinary weeks, which confirms that the flagged points are predominantly real demand surges rather than corrupted records. The remaining non-holiday outliers were inspected and found to be physically plausible sales figures with no sign of data-entry error. No rows were therefore deleted or capped. Instead, the model is supplied with holiday and calendar features so that it can learn to anticipate these surges, which is the operationally useful outcome; discarding them would have removed precisely the weeks that inventory planning most needs to get right.

## 6.5 Data Quality Before and After Processing

Table 4. Data Quality Before and After Cleaning and Feature Engineering

| Metric               | Raw dataset     | After cleaning and feature engineering                   |
|----------------------|-----------------|----------------------------------------------------------|
| Rows                 | 6,435           | 4,095 (2,340 removed — insufficient 52-week lag history) |
| Columns              | 8               | 19 (after engineered features were added)                |
| Duplicate rows       | 0               | 0                                                        |
| Missing values       | 0               | 0 (rows lacking full lag history removed, not imputed)   |
| Irregular time gaps  | 0               | 0                                                        |
| Date column dtype    | string / object | datetime64                                               |
| Outlier rows dropped | —               | 0 (retained as genuine demand signal)                    |

## 7. DATA PREPROCESSING

### 7.1 Treatment of the Store Identifier

The Store column is a numeric identifier rather than a quantity, so its integer values carry no ordinal meaning. It was retained in its numeric form rather than one-hot encoded, because the model used in this project is a tree ensemble: a decision tree can isolate any individual store through a sequence of binary splits on the identifier, so one-hot encoding would add 44 sparse columns without adding information. This choice would not be appropriate for a linear model, where the numeric identifier would wrongly imply that store 45 is nine times store 5, and Section 17.3 notes it as a point to revisit if a linear or distance-based model is introduced.

### 7.2 Feature Scaling and Transformation

Random Forests are invariant to monotonic rescaling of individual features, because each split is chosen by threshold on a single variable, so no standardisation or normalisation was applied. Two transformations were used instead, both chosen for representational rather than numerical reasons. The first is cyclical encoding of the calendar: raw month and week numbers place December and January at opposite ends of a linear scale even though they are adjacent in time, so each was mapped to a sine and cosine pair that preserves this adjacency. The second is the construction of lagged and rolling sales terms described in Section 10, which convert the raw series into predictors that carry demand history without exposing the current week.

### 7.3 Chronological Train-Test Split

Because weekly sales is a time-ordered forecasting target, a random train/test split would leak future information into training in two distinct ways: the model would see weeks that occur after the evaluation period, and the rolling-mean features of a test row could be computed from weeks that are themselves in the training set. An out-of-time split was therefore used. The final 26 weeks of every store's own history were reserved as the held-out test set, with all earlier weeks forming the training set. Applying this rule to the 4,095 model-ready rows produced 2,925 training rows covering 4 February 2011 to 27 April 2012, and 1,170 test rows covering 4 May 2012 to 26 October 2012, a 71.4% / 28.6% split with no date overlap, which was verified programmatically. Figure 2 shows the resulting timeline.



Figure 2. Out-of-Time Split Across Each Store's 143-Week History

This design is deliberately demanding. The model is trained on weeks that include the Thanksgiving and Christmas peaks of 2011 and is then evaluated on a six-month block of 2012 that it has never seen, including the summer trough and the early autumn recovery. Because the test window falls at the end of every store's series simultaneously, the evaluation reflects genuine forward-looking prediction rather than interpolation between known points.

## 7.4 Final Feature Set

Thirteen features were supplied to the model, grouped as follows:

- Historical sales: Sales\_Lag\_1, Sales\_Lag\_52, Sales\_Roll\_Mean\_4
- Calendar and seasonality: Month\_Sin, Month\_Cos, Week\_Sin, Week\_Cos
- Known operational information: Store, Holiday\_Flag
- Macroeconomic and environmental context: Temperature, Fuel\_Price, CPI, Unemployment

The target is Weekly\_Sales. Every feature in this set is either known in advance (calendar position, holiday schedule, store identity) or derived exclusively from weeks strictly earlier than the one being predicted, which is what makes the evaluation in Section 12 a fair estimate of forward-looking performance.

## 8. EXPLORATORY DATA ANALYSIS

Exploratory analysis focused on the questions that matter for inventory planning: how sales move over time, how much stores differ from one another, how much holidays lift demand, and whether the macroeconomic covariates carry usable signal.

### 8.1 Aggregate Sales Over Time



Figure 3. Aggregate Weekly Sales Across All Stores (Red Lines Mark Holiday Weeks)

Total network sales hold a stable baseline of roughly \$45–\$50 million per week for most of the year, punctuated by two dramatic and almost identically shaped spikes in late November and late December of both 2010 and 2011, where aggregate sales reach approximately \$66 million and \$81 million respectively. Each peak is immediately followed by a sharp post-holiday trough in early January. The repetition of this pattern across two consecutive years is the single most important observation in the entire exploratory analysis: it is what makes a 52-week lag a credible predictor, and it anticipates the feature-importance result reported in Section 12.4.

### 8.2 Store Volatility Segmentation

Stores were segmented by coefficient of variation (the ratio of the standard deviation of weekly sales to its mean), which normalises volatility across stores that trade at very different absolute levels. The 45 stores were divided into three equally sized tiers of 15 stores each using quantile cuts.

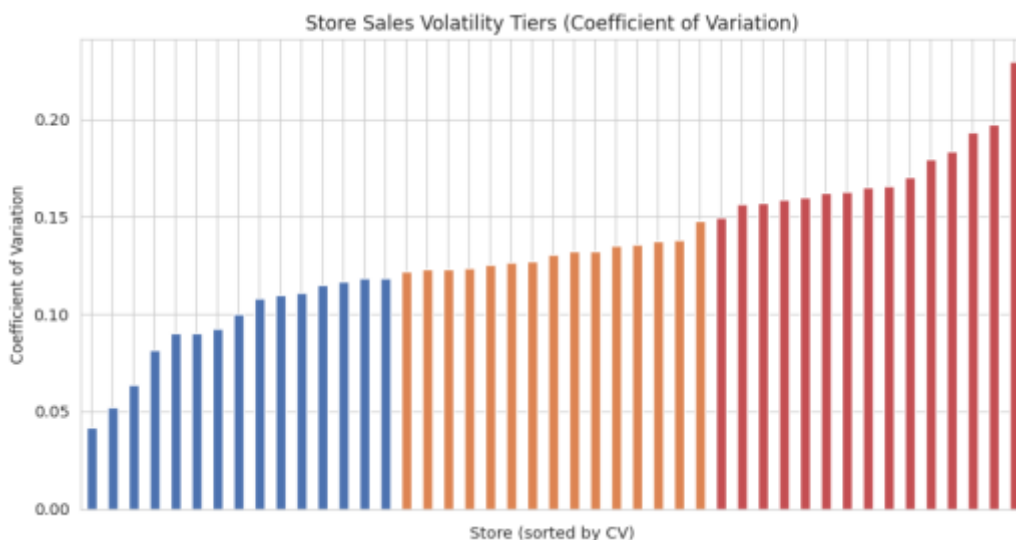


Figure 4. Store Sales Volatility Tiers by Coefficient of Variation

Volatility varies more than fivefold across the network, from approximately 0.04 at the most stable store to 0.23 at the most erratic. Store 35 is the most volatile (CV 0.230), followed by stores 7 (0.197) and 15 (0.193); the ten most volatile stores are listed in Appendix B, Table 11. Crucially, volatility is not simply a function of size: store 23 trades at a mean of \$1.39 million per week and store 36 at \$0.37 million, yet both sit in the high-volatility tier. This means a safety-stock policy keyed to store revenue would misallocate buffer inventory, a point developed in the recommendations in Section 17.1.

### 8.3 Holiday Effect



Figure 5. Weekly Sales Distribution: Holiday versus Non-Holiday Weeks

Holiday weeks shift the sales distribution upward and widen it. The median rises from \$956,211 to \$1,018,538, an increase of 6.52%, while the upper quartile rises from \$1.42 million to \$1.56 million. The dispersion also grows: the standard deviation is 12.3% higher in holiday weeks. Both distributions retain a long right tail, but the non-holiday tail is populated by a larger number of individually flagged extreme weeks, consistent with the anomaly profile in Section 6.4. The formal significance test of this difference is reported in Section 9.2.

## 8.4 Correlation With Macroeconomic Indicators

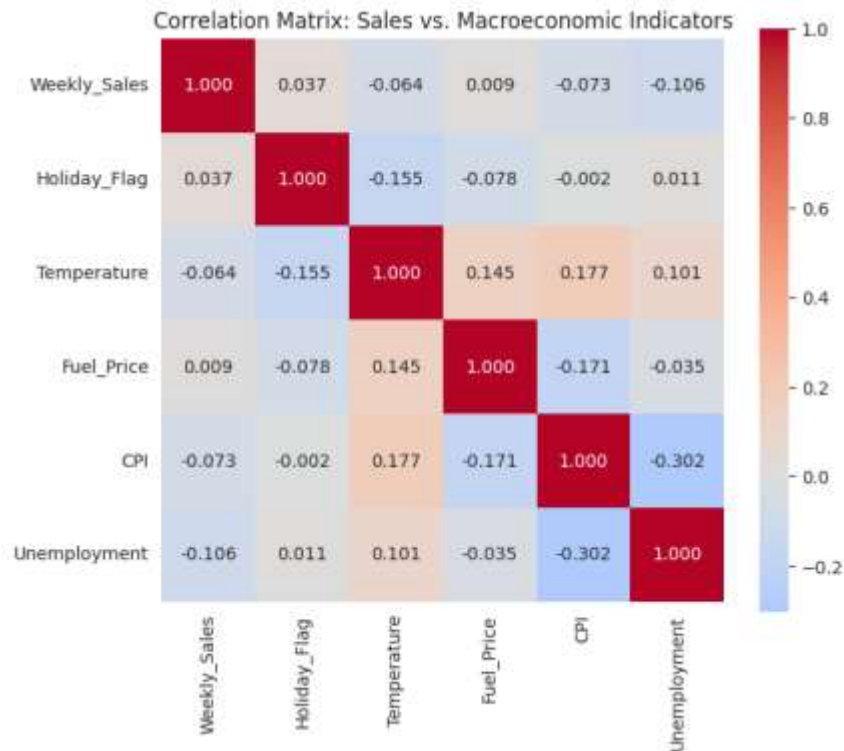


Figure 6. Correlation Matrix: Sales Versus Macroeconomic Indicators

At the pooled panel level, every macroeconomic and environmental variable is only weakly associated with weekly sales. Unemployment is the strongest of them at  $r = -0.106$ , followed by CPI at  $-0.073$ , temperature at  $-0.064$ , the holiday flag at  $0.037$ , and fuel price at  $0.009$ . Squaring these values puts the result in perspective: unemployment explains about 1.1% of the raw variance in weekly sales and the holiday flag about 0.14%. The matrix also shows a moderate negative relationship between CPI and unemployment ( $r = -0.302$ ) and mild associations among the environmental variables, which is economically sensible but says nothing about sales.

These weak pooled correlations should not be read as evidence that the covariates are meaningless. Weekly sales in this panel are dominated by store identity and by seasonality, both of which vary far faster and more widely than macroeconomic series that move slowly over months. A correlation computed across all stores and all weeks simultaneously therefore averages out whatever within-store effect these variables have. The practical implication for modelling is nonetheless clear, and it is confirmed by the feature-importance results in Section 12.4: these variables should not be expected to carry the forecast.

## 9. STATISTICAL ANALYSIS

### 9.1 Descriptive Statistics

Table 5. Descriptive Statistics of Key Variables

| Variable              | Mean         | Std Dev    | Min        | Median     | Max          |
|-----------------------|--------------|------------|------------|------------|--------------|
| Weekly_Sales (target) | 1,046,964.88 | 564,366.62 | 209,986.25 | 960,746.04 | 3,818,686.45 |
| Holiday_Flag          | 0.07         | 0.26       | 0.00       | 0.00       | 1.00         |
| Temperature (°F)      | 60.66        | 18.44      | -2.06      | 62.67      | 100.14       |
| Fuel_Price (USD/gal)  | 3.36         | 0.46       | 2.47       | 3.44       | 4.47         |
| CPI                   | 171.58       | 39.36      | 126.06     | 182.62     | 227.23       |
| Unemployment (%)      | 8.00         | 1.88       | 3.88       | 7.87       | 14.31        |

Weekly sales average \$1.05 million with a standard deviation of \$564,367, and the mean sits above the median of \$960,746, indicating a right-skewed distribution driven by the holiday peaks and by the largest stores. The range is wide: the quietest store-week records \$209,986 while the busiest records \$3.82 million, a factor of more than eighteen. The covariates span economically meaningful ranges — unemployment from 3.88% to 14.31%, CPI from 126.06 to 227.23, fuel from \$2.47 to \$4.47 per gallon — which confirms that their weak correlation with sales reflects a genuinely weak relationship rather than insufficient variation.

### 9.2 Holiday Effect: Welch's t-test

To test whether the holiday gap visible in Figure 5 is statistically real rather than sampling noise, a Welch's independent-samples t-test was run comparing weekly sales in holiday and non-holiday weeks. Welch's form was chosen over the pooled-variance form precisely because the two groups have unequal variances and very unequal sizes (450 against 5,985), conditions under which the pooled test is unreliable. Table 6 summarises the two groups and the test result.

Table 6. Holiday versus Non-Holiday Weekly Sales

| Statistic               | Non-holiday weeks | Holiday weeks         |
|-------------------------|-------------------|-----------------------|
| Count                   | 5,985             | 450                   |
| Mean                    | \$1,041,256.38    | \$1,122,887.89        |
| Median                  | \$956,211.20      | \$1,018,538.04        |
| Standard deviation      | \$558,957.44      | \$627,684.93          |
| Mean difference         | —                 | +\$81,631.51 (+7.84%) |
| Median difference       | —                 | +\$62,326.84 (+6.52%) |
| Welch's t-statistic     | —                 | 2.680                 |
| p-value                 | —                 | 0.0076                |
| 95% confidence interval | —                 | \$21,790 to \$141,473 |
| Cohen's d               | —                 | 0.14 (small effect)   |

The result is statistically significant at the 5% level ( $p = 0.0076$ ), so the holiday lift is real. Its practical size must nonetheless be stated honestly. The mean difference of \$81,632 per store-week is commercially substantial, and the 95% confidence interval excludes zero by a wide margin, but Cohen's  $d$  of 0.14 is a small standardised effect: the two distributions overlap heavily, and a randomly chosen holiday week is only modestly more likely to outsell a randomly chosen ordinary week. The reason both statements are true simultaneously is that store-to-store variation in this panel is enormous relative to the holiday effect. This is also why aggregation matters operationally: pooled across 45 stores the holiday uplift is worth several million dollars of network revenue per holiday week, even though it is modest relative to the noise at any individual store.

### 9.3 Interpretation and Caveats

Two cautions should be carried forward into the discussion. First, the holiday flag marks the week containing a holiday, not the trading pattern around it; Figure 3 shows that the largest spikes occur in the weeks immediately preceding Christmas, which are not all flagged, so the flag understates the true seasonal effect and a richer holiday-proximity feature would likely perform better. Second, all correlations and group differences reported here are associations observed in a single 143-week window; none of them establishes causation, and none of them can be confirmed as stable across a longer horizon from this dataset alone.

## 10. FEATURE ENGINEERING

Because the dataset is a collection of 45 weekly time series rather than an independent table, most of the engineering effort went into constructing predictors that use only information genuinely available before the week being forecast. All lag and rolling features were computed after sorting by Store and Date and grouping by Store, so no value ever crosses from one store's history into another's, and every rolling statistic was shifted by one week before the window was applied, so the week being predicted never contributes to its own feature. Table 7 documents the resulting features.

**Table 7. Engineered Features**

| Feature              | Description and logic                                                                                                                                                                       |
|----------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Year, Week, Month    | ISO calendar components extracted from Date                                                                                                                                                 |
| Month_Sin, Month_Cos | Sine and cosine encoding of month on a 12-period cycle, so December and January are numerically adjacent rather than 11 units apart                                                         |
| Week_Sin, Week_Cos   | Sine and cosine encoding of ISO week on a 52-period cycle, giving the model a smooth representation of position within the trading year                                                     |
| Sales_Lag_1          | The same store's sales one week earlier — captures short-term momentum and the current trading level                                                                                        |
| Sales_Lag_52         | The same store's sales in the same week of the previous year — captures annual seasonality, including holiday peaks                                                                         |
| Sales_Roll_Mean_4    | Four-week trailing mean of prior sales, computed with shift(1) before rolling so the current week is excluded — a smoothed measure of the recent level that is less noisy than a single lag |

Constructing these features necessarily introduced missing values at the start of each store's series: 45 rows for the one-week lag (one per store), 180 for the four-week rolling mean, and 2,340 for the 52-week lag, the last being the binding constraint at 52 rows per store. These rows were removed rather than imputed. Imputing a prior-year sales value for a period in which no prior year exists would mean inventing the exact signal the model is intended to learn from, and would inflate the reported accuracy. The cost of this decision is stated plainly: 2,340 rows, or 36.4% of the raw panel, are consumed as warm-up history, leaving 4,095 rows and 19 columns in the final modelling dataset. The benefit is that every remaining row carries a complete, genuine feature vector.

## 11. MODEL DEVELOPMENT

### 11.1 Problem Formulation

This is a supervised regression problem. The target, `Weekly_Sales`, is a continuous monetary quantity, not a class label, so the task is to minimise expected prediction error rather than to maximise classification accuracy. The inputs are the thirteen leakage-safe features defined in Section 7.4, and the evaluation is performed entirely on the chronologically separated test period defined in Section 7.3.

### 11.2 Model Selection

A Random Forest regressor was selected as the primary model for four reasons that follow directly from the structure of this dataset. It captures non-linear relationships and interactions between features without those interactions having to be specified in advance, which matters because the effect of a holiday week plausibly depends on the store and the season. It handles features on completely different scales — dollars, degrees Fahrenheit, index points, percentages — without any rescaling. It treats the numeric Store identifier sensibly, as discussed in Section 7.1. And it provides a direct feature-importance measure, which makes the resulting forecast explainable to a planning team rather than opaque.

Two naive baselines were evaluated on the identical test rows so that the value added by the model can be measured rather than asserted. The first predicts this week's sales as last week's sales (`Sales_Lag_1`), representing a planner who simply repeats the most recent observation. The second predicts this week's sales as the same week last year (`Sales_Lag_52`), representing a planner who works from the previous year's calendar. Both are genuinely competitive heuristics in retail, which makes them a meaningful bar to clear.

### 11.3 Hyperparameter Tuning

Hyperparameters were tuned with `RandomizedSearchCV` using three-fold cross-validation and negative mean absolute error as the scoring criterion. Fifteen candidate configurations were sampled from a grid of 540 possible combinations, giving 45 model fits in total. Randomised search was preferred over exhaustive grid search because it explores a far wider region of the space for the same computational budget, and MAE was preferred over the default  $R^2$  because it expresses error in dollars, which is the unit in which an inventory decision is actually taken. Critically, the search was fitted on the training partition only, so the held-out test weeks played no part in model selection. Table 8 reports the search space and the selected configuration.

**Table 8. Hyperparameter Search Space and Selected Configuration**

| Hyperparameter                 | Values searched                          | Selected                 |
|--------------------------------|------------------------------------------|--------------------------|
| <code>n_estimators</code>      | 200, 300, 400, 500                       | 500                      |
| <code>max_depth</code>         | None, 10, 15, 20, 25                     | None (unrestricted)      |
| <code>min_samples_split</code> | 2, 5, 10                                 | 2                        |
| <code>min_samples_leaf</code>  | 1, 2, 4                                  | 1                        |
| <code>max_features</code>      | 'sqrt', 0.5, 1.0                         | 1.0 (all features)       |
| Search strategy                | 15 of 540 combinations (2.8%), 3-fold CV | 45 fits total            |
| Scoring criterion              | Negative mean absolute error             | Best CV MAE: \$59,153.23 |

The selected configuration is notable for being maximally flexible: unrestricted depth, the smallest permitted leaf and split sizes, and all features considered at every split. In many settings this combination would be a

warning sign of overfitting. Here it is defensible, because the dominant predictors are lagged sales values whose relationship to the target is close to identity, and because the out-of-time test results in Section 12 confirm that the fitted model generalises to a period it never saw. The best cross-validated MAE of \$59,153 is itself discussed in Section 12.6, where it is compared against the test-period figure.

## 11.4 Training the Model

The core training and prediction code was as follows; the complete, ordered, and runnable code is provided in Appendix A and in the accompanying Jupyter notebook.

```
rf = RandomForestRegressor(random_state=42, n_jobs=-1)

search = RandomizedSearchCV(
    estimator=rf,
    param_distributions=param_distributions,
    n_iter=15, cv=3, scoring='neg_mean_absolute_error',
    random_state=42, n_jobs=-1, verbose=1)

search.fit(X_train, y_train)

best_model = search.best_estimator_
test_pred = best_model.predict(X_test)
```

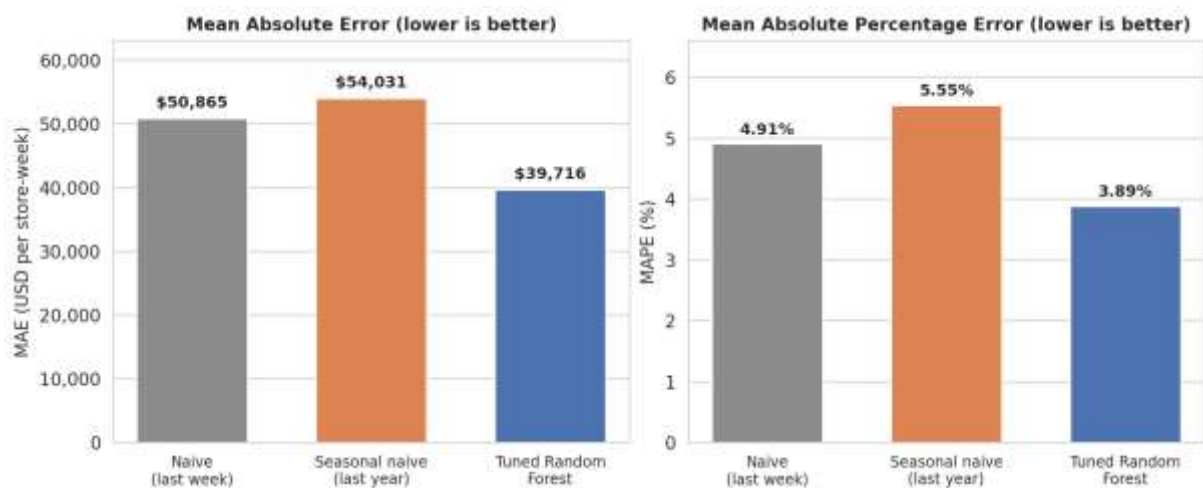
## 12. MODEL EVALUATION

### 12.1 Regression Metrics

All models were evaluated on the identical held-out test period (4 May 2012 to 26 October 2012, 1,170 store-week records) using four complementary metrics. Mean absolute error reports the average error in dollars; root mean squared error penalises large errors more heavily and therefore exposes occasional bad misses; mean absolute percentage error expresses the error relative to each store's own sales level, which makes small and large stores comparable; and  $R^2$  reports the share of test-period variance explained. Table 9 ranks the tuned model against both baselines.

**Table 9. Test-Period Performance Against Naive Baselines**

| Model                      | MAE (USD) | MAPE (%) | Improvement in MAE |
|----------------------------|-----------|----------|--------------------|
| Tuned Random Forest        | 39,715.89 | 3.89     | —                  |
| Naive (last week)          | 50,864.96 | 4.91     | 21.92% worse       |
| Seasonal naive (last year) | 54,031.40 | 5.55     | 26.49% worse       |



**Figure 7. Tuned Model Versus Naive Baselines on the Held-Out Test Period**

The tuned Random Forest achieves a test-period MAE of \$39,715.89, an RMSE of \$62,910.81, a MAPE of 3.89%, and an  $R^2$  of 0.9862. It reduces MAE by 21.92% against the last-week baseline and by 26.49% against the last-year baseline, and it improves MAPE by 1.02 and 1.66 percentage points respectively. Expressed in the terms a planner would use, the model is typically wrong by about 3.8% of an average store-week's revenue, and switching from the last-week heuristic to the model removes roughly \$11,000 of average error per store-week.

The ordering of the two baselines is itself informative: the last-week heuristic outperforms the last-year heuristic by 6.2% on MAE, which shows that recent trading level carries more short-horizon information than the prior-year calendar taken alone. The model beats both because it combines them, a reading confirmed by the feature-importance analysis in Section 12.4.

### 12.2 Classification Evaluation

Not applicable. This project is formulated as a regression task predicting a continuous revenue quantity, so no confusion matrix, accuracy, precision, recall, or F1-score is reported. The equivalent diagnostic role is played by the residual analysis in Section 12.5.

## 12.3 Visual Validation

Aggregate metrics can conceal systematic failures, so a single store's held-out trajectory was inspected directly. Figure 8 plots actual against predicted weekly sales for Store 1 across the full 26-week test window.



Figure 8. Store 1 — Actual Versus Predicted Weekly Sales, Held-Out Period

The predicted series tracks the level and the turning points of the actual series closely for most of the window, and the two curves are near-coincident in early June, late July, late August, and the first week of October. Two characteristic behaviours are visible. The model tends to under-predict the sharpest upward spikes, most obviously the isolated peak near \$1.77 million in early July that it forecasts at roughly \$1.57 million; this is the expected conservatism of an averaging ensemble faced with a one-off surge. It also smooths some week-to-week oscillation, following the trend rather than every fluctuation. Neither behaviour represents a systematic bias in one direction, which is confirmed by the residual analysis below.

## 12.4 Feature Importance

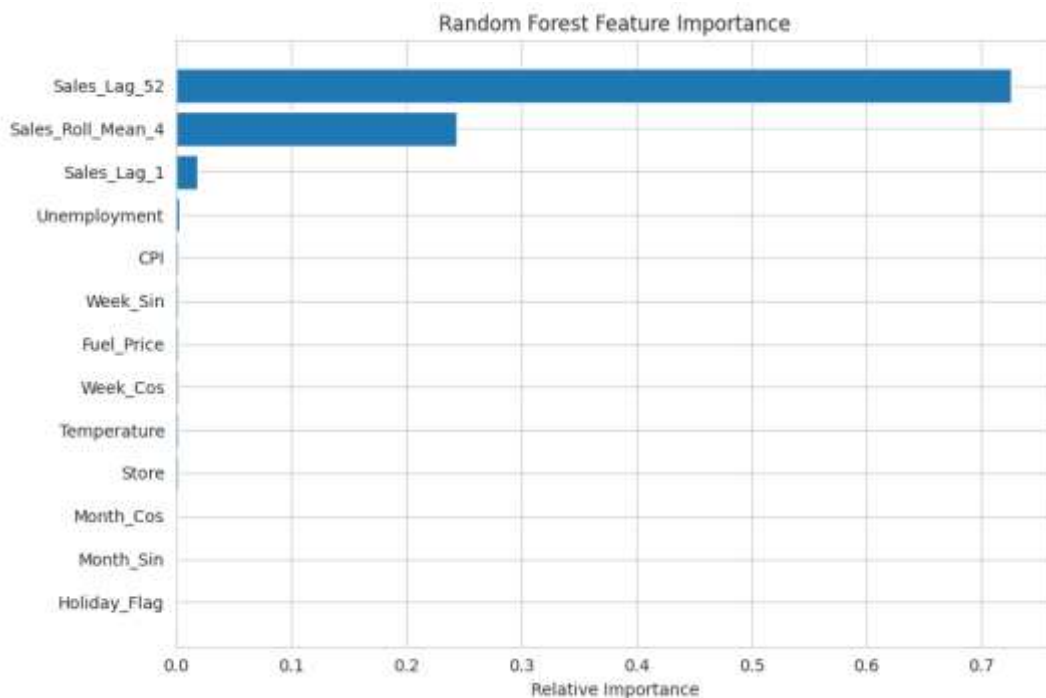


Figure 9. Random Forest Feature Importance

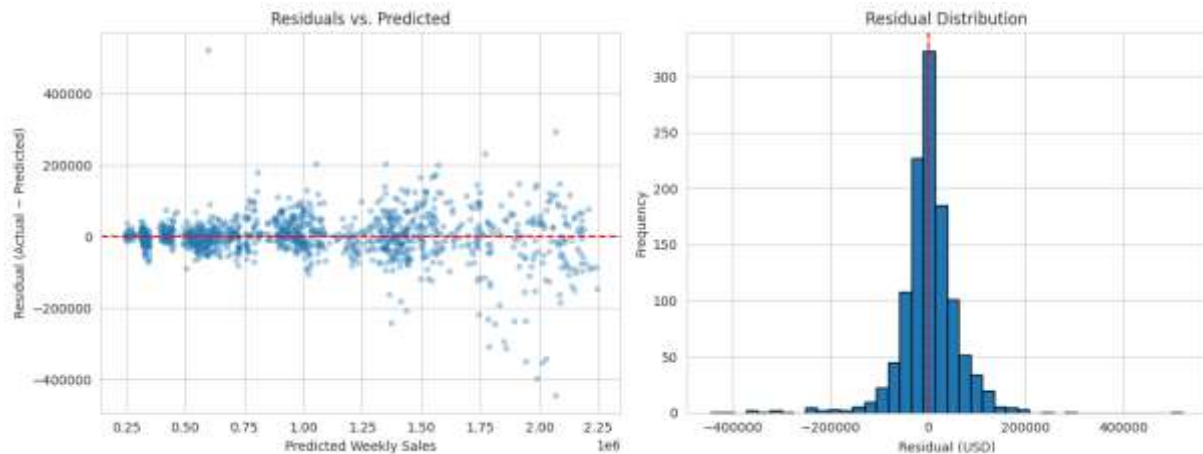
Table 10. Feature Importance Ranking

| Rank | Feature           | Importance (%) | Family              |
|------|-------------------|----------------|---------------------|
| 1    | Sales_Lag_52      | 72.60          | Annual seasonality  |
| 2    | Sales_Roll_Mean_4 | 24.40          | Recent level        |
| 3    | Sales_Lag_1       | 1.82           | Short-term momentum |
| 4    | Unemployment      | 0.28           | Macroeconomic       |
| 5    | CPI               | 0.17           | Macroeconomic       |
| 6    | Week_Sin          | 0.14           | Calendar            |
| 7    | Fuel_Price        | 0.12           | Macroeconomic       |
| 8    | Week_Cos          | 0.12           | Calendar            |
| 9    | Temperature       | 0.12           | Environmental       |
| 10   | Store             | 0.12           | Identifier          |
| 11   | Month_Cos         | 0.06           | Calendar            |
| 12   | Month_Sin         | 0.04           | Calendar            |
| 13   | Holiday_Flag      | 0.02           | Operational         |

The ranking is strikingly concentrated. Sales in the same week one year earlier alone accounts for 72.60% of predictive power, the four-week trailing mean for a further 24.40%, and the two together for 97.00%. Adding the one-week lag brings the top three to 98.82%, leaving all four macroeconomic and environmental variables with a combined 0.69% and all four calendar encodings with 0.36%. The holiday flag ranks last at 0.02%.

The apparent contradiction with Section 9.2, where the holiday effect was statistically significant, resolves cleanly once the mechanism is understood. Random Forest importance measures the marginal contribution of a feature given the others already present. Because Sales\_Lag\_52 contains the prior year's value for the same calendar week, it already encodes the holiday surge, the seasonal trough, and the store's baseline scale simultaneously. The holiday flag adds almost nothing beyond what the lag has already supplied. Its low importance therefore indicates redundancy, not irrelevance — a distinction that matters, because it would be wrong to conclude from this table that holidays do not affect sales.

## 12.5 Residual Analysis



**Figure 10. Residual Diagnostics: Residuals Versus Predicted and Residual Distribution**

Residuals are centred almost exactly on zero, with a mean of \$1,066.91 and a median of \$2,050.36 against a mean sales level of \$1.05 million, which is a bias of roughly one tenth of one percent. The residual distribution is sharply peaked and approximately symmetric, with a standard deviation of \$62,901.76. The residuals versus predicted plot shows a clear funnel shape: the spread of errors widens as predicted sales increase, which is classic heteroscedasticity and is expected here, since a large store simply has more dollars available to be wrong about. In proportional terms the error is far more stable, which is why the MAPE of 3.89% is the more meaningful headline figure. The largest individual misses, exceeding \$400,000, are concentrated among the highest-volume store-weeks and are consistent with the spike under-prediction visible in Figure 8. The ratio of RMSE to MAE is 1.58, moderately above the value of about 1.25 expected for normally distributed errors, which quantifies the influence of that small number of large misses.

## 12.6 Overfitting and Generalisation

Because the selected configuration places no limit on tree depth, the risk of overfitting deserves explicit attention. The evidence available points against it. The best cross-validated MAE obtained during tuning was \$59,153.23, whereas the MAE on the genuinely unseen test period was \$39,715.89 — the model performs 48.9% better on the held-out future than on its own cross-validation folds. A model that had memorised its training data would show the opposite pattern.

The explanation lies in the structure of the cross-validation rather than in any property of the model. RandomizedSearchCV uses standard k-fold splitting, which partitions the training rows without regard to time and without regard to store, so each fold trains on a fragmented subset of each store's history and is scored on weeks drawn from throughout that history, including the volatile November and December peaks of 2011. The test period, by contrast, is a contiguous six-month block from May to October 2012 that contains no major holiday peak and is therefore intrinsically easier to predict. The cross-validation score is best understood as a conservative internal ranking signal used to choose between configurations, not as an estimate of deployed accuracy. It should also be noted that training-set metrics were not computed in the notebook, so the conventional train-versus-test comparison is unavailable; this is recorded as a limitation in Section 15 and as a first item for future work in Section 17.3.

## 13. RESULTS AND FINDINGS

- **Annual seasonality dominates weekly sales.** Sales in the same week one year earlier account for 72.60% of the tuned model's predictive power, and together with the four-week trailing mean explain 97.00% of it (Table 10, Figure 9). The aggregate sales curve confirms the mechanism visually: the November and December peaks of 2010 and 2011 are almost identical in shape and timing (Figure 3).
- **The panel is complete and structurally sound.** All 45 stores contribute exactly 143 consecutive weekly records with no duplicates, no missing values, and no irregular gaps, so no imputation or calendar reconstruction was required (Sections 6.1 to 6.3).
- **Holiday weeks produce a statistically significant but standardised-small lift.** Mean sales rise 7.84% and median sales 6.52% in holiday weeks, a mean difference of \$81,632 per store-week (Welch's  $t = 2.680$ ,  $p = 0.0076$ , 95% CI \$21,790 to \$141,473), but Cohen's  $d$  is only 0.14 because store-to-store variation dwarfs the effect (Table 6).
- **Extreme sales weeks are genuine demand, not data errors.** Store-specific IQR profiling flagged 300 weeks (4.66%), of which holiday weeks are flagged 5.26 times as often as ordinary weeks; all rows were retained so the model could learn to anticipate surges rather than be blind to them (Table 3).
- **The tuned model clearly beats both naive baselines.** On the held-out final 26 weeks of every store's history, the Random Forest achieved MAE \$39,715.89, RMSE \$62,910.81, MAPE 3.89%, and  $R^2$  0.9862, improving on the last-week baseline by 21.92% and the last-year baseline by 26.49% on MAE (Table 9, Figure 7).
- **Macroeconomic covariates contribute almost nothing once sales history is present.** Unemployment, CPI, fuel price, and temperature together account for 0.69% of feature importance, consistent with their weak raw correlations of  $|r| \leq 0.106$  (Figure 6, Table 10).
- **Store volatility varies more than fivefold and is independent of store size.** Coefficients of variation range from roughly 0.04 to 0.23, and stores in the high-volatility tier include both the largest and among the smallest in the network (Figure 4, Appendix B Table 11).
- **Forecast errors are unbiased but heteroscedastic.** Residuals are centred near zero (mean \$1,067, about 0.1% of mean sales) yet widen with predicted sales, so error should be budgeted in proportional rather than absolute terms (Figure 10).

## 14. DISCUSSION

Taken together, the exploratory, statistical, and modelling results tell a coherent story about what actually governs weekly sales in this network. Revenue at any given store is driven overwhelmingly by two forces, both internal to the store's own history: where the week sits in the annual trading cycle, and the level at which the store has been trading recently. External economic conditions, at least as represented by the four covariates available here, are close to irrelevant once those two forces are accounted for. This is why a model given the prior year's same-week sales and a four-week trailing mean reaches an  $R^2$  of 0.9862 while assigning less than one percent of its importance to every macroeconomic and calendar variable combined.

The most instructive tension in the results is between the statistical significance of the holiday effect in Section 9.2 and the near-zero importance of the holiday flag in Section 12.4. Both findings are correct, and reconciling them is what makes the analysis useful rather than merely accurate. The t-test asks whether holiday weeks differ from ordinary weeks when nothing else is known, and the answer is yes. Feature importance asks whether the holiday flag adds information beyond the other features, and the answer is almost none — because `Sales_Lag_52` already carries last year's holiday surge for exactly that calendar week. The practical lesson for a planning team is that the value of a holiday indicator depends entirely on what else is in the model: in a system with reliable year-over-year history it is redundant, but for a new store with no prior-year data it would be one of the few signals available.

The performance figures deserve to be read with the same care. An  $R^2$  of 0.9862 is high in absolute terms, but a substantial part of that explained variance comes from the model correctly distinguishing a \$1.4 million store from a \$0.4 million store, which the lag features make almost trivial. The more honest measure of forecasting skill is the comparison against baselines that also have access to lagged sales, and on that measure the model delivers a genuine but bounded improvement: roughly 22% lower MAE than simply repeating last week. That is a meaningful operational gain, not a transformation, and it should be presented to stakeholders as such.

Finally, the volatility analysis points to an actionable insight that the headline accuracy metrics obscure entirely. Because errors scale with store size while volatility does not, a single network-wide safety-stock rule would systematically over-protect stable large stores and under-protect erratic small ones. Segmenting replenishment policy by the volatility tiers established in Section 8.2 addresses this directly, and does so using nothing more than the descriptive statistics already computed.

## 15. LIMITATIONS

- **Short historical window.** The panel covers 143 weeks, which provides only two complete observations of each annual seasonal cycle. Because the 52-week lag consumes the first year entirely, the model effectively learns seasonality from a single prior year, and that pattern cannot be confirmed as stable over a longer horizon from this dataset.
- **Substantial data consumed as warm-up.** Requiring a 52-week lag removed 2,340 of 6,435 rows, or 36.4% of the panel, before modelling began. A shorter-memory model would have retained more data, and the trade-off between lag depth and sample size was not formally evaluated.
- **Single test window, no walk-forward validation.** Performance is measured on one contiguous six-month block (May to October 2012) that contains no major holiday peak. Rolling-origin validation across several windows, including at least one holiday season, would give a more representative estimate.
- **Cross-validation was not time-aware.** RandomizedSearchCV used standard k-fold splitting, which breaks the temporal ordering within the training period. TimeSeriesSplit would have been the methodologically correct choice and would have produced a cross-validation score directly comparable to the test result.
- **Training-set metrics were not computed.** Without training-period MAE and  $R^2$ , the conventional train-versus-test overfitting diagnostic is unavailable, and the generalisation argument in Section 12.6 rests on the cross-validation comparison alone.
- **One-week-ahead horizon only.** The formulation predicts a single week ahead using the immediately preceding week. Real replenishment decisions often require forecasts several weeks out, which would need either recursive prediction or a direct multi-horizon model, and accuracy would be expected to degrade with horizon.
- **Holiday information is coarse.** A binary flag marks only the week containing a holiday. It does not distinguish Thanksgiving from Labor Day, nor capture the pre-holiday build-up weeks that Figure 3 shows to be the largest peaks of the year.
- **No store-level context.** The dataset contains no store size, format, region, department mix, promotional calendar, or local competition data, all of which plausibly explain part of the store-to-store variation in both level and volatility.
- **Associations, not causal effects.** Every relationship reported here is observational. The holiday lift, the macroeconomic correlations, and the volatility tiers describe patterns in the data and should not be interpreted as causal levers.

## 16. CONCLUSION

This project set out to determine how accurately weekly sales at an individual Walmart store can be forecast one week ahead using only information available at forecast time, and how much a tuned machine learning model improves on the heuristics a planner could apply without one. The panel of 6,435 store-week records proved to be complete and structurally sound, requiring no imputation and no de-duplication, and a store-specific anomaly profile established that its extreme weeks are genuine demand surges rather than defects — a finding that justified retaining every row.

A leakage-aware feature set of cyclical calendar encodings, one-week and fifty-two-week sales lags, and a four-week trailing mean was constructed, and a Random Forest regressor was tuned by randomised search on the training period alone. Evaluated on the final 26 weeks of every store's history, a period the model had never seen, it achieved a mean absolute error of \$39,715.89, a mean absolute percentage error of 3.89%, and an  $R^2$  of 0.9862, improving on a last-week naive baseline by 21.92% and a last-year seasonal baseline by 26.49%. Residuals were unbiased, centred within one tenth of one percent of zero, though their spread widened with store size.

The substantive finding is that weekly sales in this network are governed almost entirely by a store's own recent and year-ago history: the 52-week lag and the four-week rolling mean together carry 97.00% of the model's predictive power, while the macroeconomic covariates contribute less than one percent. The holiday effect is statistically real yet adds little once year-ago sales are known, because that lag already encodes it. The practical consequence is that inventory planning for this network should be anchored on year-over-year tracking adjusted by the recent four-week trend, with replenishment policy segmented by the store volatility tiers identified here rather than by store revenue. These recommendations, and the steps required to move the model from notebook to production, are set out in Section 17.

## 17. RECOMMENDATIONS AND FUTURE WORK

### 17.1 Operational Recommendations

- **Anchor inventory plans on 52-week lag tracking.** Because year-ago same-week sales carry 72.60% of predictive power, store-level plans should start from what that week delivered last year and then be adjusted by the recent four-week trend, rather than being built from macroeconomic outlooks.
- **Segment safety stock by volatility tier, not by store revenue.** Coefficients of variation span roughly 0.04 to 0.23 independently of store size, so the 15 high-volatility stores identified in Section 8.2 warrant larger buffers and more frequent review cycles than equally large but stable stores.
- **Pre-position inventory for the November–December peak specifically.** Figure 3 shows aggregate sales rising from a \$45–50 million baseline to approximately \$81 million in the final December week of 2011, with the steepest build-up in the weeks before the flagged holiday itself. Replenishment should lead the flagged week rather than coincide with it.
- **Budget forecast error proportionally.** Residual spread grows with store size while proportional error stays near 3.89%, so buffer sizing and exception thresholds should be expressed as a percentage of expected sales rather than as a single dollar tolerance across the network.
- **Use the model as a decision aid with human review on spikes.** The model under-predicts isolated surges (Figure 8), so weeks with an unusually large year-over-year change should be routed for manual review rather than actioned automatically.
- **Treat new or refitted stores as a separate case.** Any store without 52 weeks of history cannot be served by the primary model at all. Such stores need a fallback that leans on the holiday flag, calendar features, and comparable-store averages until their own history accumulates.

### 17.2 Deployment Roadmap

Moving from a static notebook to a production forecasting service would involve four components:

1. **Experiment tracking.** Log every search run, its hyperparameters, and its MAE, RMSE, MAPE, and  $R^2$  with a tool such as MLflow, so that model versions remain comparable and reproducible.
2. **Scheduled orchestration.** Run a weekly pipeline that ingests the latest sales and macroeconomic data, regenerates the lag and rolling features, and produces the next week's forecast for all 45 stores.
3. **Batch inference to a warehouse table.** Write scored forecasts to a table that inventory and staffing systems read directly, rather than exporting spreadsheets by hand.
4. **Monitoring and retraining triggers.** Track live MAE and MAPE against the values established in this report, and alert when error drifts materially above the 3.89% test-period MAPE so that an unmodelled shock is caught quickly.

### 17.3 Future Work

Several extensions follow directly from the limitations recorded in Section 15. The most immediate is to replace k-fold cross-validation with TimeSeriesSplit and to compute training-set metrics, which together would make the generalisation argument complete rather than inferential. Beyond that, walk-forward validation across multiple origins including at least one holiday season would give a more representative accuracy estimate; richer holiday features distinguishing individual holidays and encoding weeks-until-holiday would likely capture the pre-Christmas build-up that the binary flag misses; gradient-boosted alternatives such as XGBoost or LightGBM and classical seasonal models such as SARIMA or Prophet would establish whether the Random Forest result is method-specific; direct multi-horizon forecasting would serve

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replenishment lead times longer than one week; and the addition of store attributes, department mix, and promotional calendars would address the largest remaining gap in the feature set.

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## APPENDIX A. CODE

The excerpts below are the key steps referenced throughout the report. The complete, ordered, and runnable code is provided in the accompanying Jupyter notebook (Walmart\_Sales\_Forecasting\_Capstone.ipynb).

### A.1 Data Loading and Profiling

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats

df = pd.read_csv('P_6_Walmart.csv')
df.info()
df.describe().T
```

### A.2 Data Quality and Panel Continuity Audit

```
print('Duplicate rows:', df.duplicated().sum())
df.isna().sum()

df['Date'] = pd.to_datetime(df['Date'], format='%d-%m-%Y')
df = df.sort_values(['Store', 'Date']).reset_index(drop=True)

# every store must advance in strict 7-day steps
gap_deltas = (df.groupby('Store')['Date']
               .apply(lambda s: s.diff().dropna().unique()))
all_gaps = set()
for arr in gap_deltas:
    all_gaps.update(arr)
assert all_gaps == {pd.Timedelta(days=7)}
```

### A.3 Store-Specific Anomaly Profiling

```
def flag_outliers(group):
    q1, q3 = group['Weekly_Sales'].quantile([0.25, 0.75])
    iqr = q3 - q1
    lower, upper = q1 - 1.5 * iqr, q3 + 1.5 * iqr
    return (group['Weekly_Sales'] < lower) | (group['Weekly_Sales'] > upper)

df['Is_Outlier'] = (df.groupby('Store', group_keys=False)
                    .apply(flag_outliers, include_groups=False))
```

## A.4 Feature Engineering (Leakage-Safe)

```
df['Year'] = df['Date'].dt.isocalendar().year.astype(int)
df['Week'] = df['Date'].dt.isocalendar().week.astype(int)
df['Month'] = df['Date'].dt.month

df['Month_Sin'] = np.sin(2 * np.pi * df['Month'] / 12)
df['Month_Cos'] = np.cos(2 * np.pi * df['Month'] / 12)
df['Week_Sin'] = np.sin(2 * np.pi * df['Week'] / 52)
df['Week_Cos'] = np.cos(2 * np.pi * df['Week'] / 52)

grouped_sales = df.groupby('Store')['Weekly_Sales']
df['Sales_Lag_1'] = grouped_sales.shift(1)
df['Sales_Lag_52'] = grouped_sales.shift(52)
df['Sales_Roll_Mean_4'] = (
    df.groupby('Store')['Weekly_Sales']
    .transform(lambda x: x.shift(1).rolling(window=4).mean()))
```

## A.5 Statistical Testing

```
holiday_sales = df.loc[df['Holiday_Flag'] == 1, 'Weekly_Sales']
non_holiday_sales = df.loc[df['Holiday_Flag'] == 0, 'Weekly_Sales']

t_stat, p_value = stats.ttest_ind(holiday_sales, non_holiday_sales,
                                   equal_var=False)

corr_cols = ['Weekly_Sales', 'Holiday_Flag', 'Temperature',
             'Fuel_Price', 'CPI', 'Unemployment']
corr_matrix = df[corr_cols].corr()
```

## A.6 Chronological Split, Tuning and Evaluation

```
df_model = df.dropna(subset=['Sales_Lag_1', 'Sales_Lag_52',
                              'Sales_Roll_Mean_4']).copy()
df_model['Weeks_From_End'] = (df_model.groupby('Store')['Date']
                              .rank(method='first', ascending=False))

TEST_WEEKS = 26
train_df = df_model[df_model['Weeks_From_End'] > TEST_WEEKS].copy()
test_df = df_model[df_model['Weeks_From_End'] <= TEST_WEEKS].copy()

param_distributions = {
    'n_estimators': [200, 300, 400, 500],
    'max_depth': [None, 10, 15, 20, 25],
    'min_samples_split': [2, 5, 10],
    'min_samples_leaf': [1, 2, 4],
    'max_features': ['sqrt', 0.5, 1.0]}

search = RandomizedSearchCV(RandomForestRegressor(random_state=42, n_jobs=-1),
                             param_distributions, n_iter=15, cv=3,
                             scoring='neg_mean_absolute_error',
                             random_state=42, n_jobs=-1)
search.fit(X_train, y_train)

test_pred = search.best_estimator_.predict(X_test)
mae = mean_absolute_error(y_test, test_pred)
rmse = np.sqrt(mean_squared_error(y_test, test_pred))
mape = np.mean(np.abs((y_test - test_pred) / y_test)) * 100
r2 = r2_score(y_test, test_pred)
```

## APPENDIX B. ADDITIONAL FIGURES AND TABLES

**Table 11. Ten Most Volatile Stores by Coefficient of Variation**

| Store | Mean weekly sales (USD) | Std dev (USD) | Coefficient of variation |
|-------|-------------------------|---------------|--------------------------|
| 35    | 919,724.98              | 211,243.46    | 0.230                    |
| 7     | 570,617.31              | 112,585.47    | 0.197                    |
| 15    | 623,312.47              | 120,538.65    | 0.193                    |
| 29    | 539,451.43              | 99,120.14     | 0.184                    |
| 23    | 1,389,864.46            | 249,788.04    | 0.180                    |
| 21    | 756,069.08              | 128,752.81    | 0.170                    |
| 45    | 785,981.41              | 130,168.53    | 0.166                    |
| 16    | 519,247.73              | 85,769.68     | 0.165                    |
| 18    | 1,084,718.42            | 176,641.51    | 0.163                    |
| 36    | 373,511.99              | 60,725.17     | 0.163                    |

All ten stores fall in the high-volatility tier. The list spans the full size range of the network, from store 36 at \$0.37 million per week to store 23 at \$1.39 million, which is the evidence for the recommendation in Section 17.1 that safety stock be keyed to volatility rather than revenue.

**Table 12. Test-Period Residual Statistics**

| Statistic           | Value (USD) | Note                                                           |
|---------------------|-------------|----------------------------------------------------------------|
| Mean residual       | 1,066.91    | 0.10% of mean weekly sales — effectively unbiased              |
| Median residual     | 2,050.36    | Close to the mean; no strong skew                              |
| Standard deviation  | 62,901.76   | Consistent with the reported RMSE of 62,910.81                 |
| Mean absolute error | 39,715.89   | 3.79% of mean weekly sales                                     |
| RMSE / MAE ratio    | 1.58        | Above the $\approx 1.25$ of normal errors — a few large misses |

Table 13. Evaluation Metrics Summary, Tuned Random Forest

| Metric         | Value               | Interpretation                                         |
|----------------|---------------------|--------------------------------------------------------|
| MAE            | \$39,715.89         | Average absolute error per store-week                  |
| RMSE           | \$62,910.81         | Penalises the large misses on high-volume weeks        |
| MAPE           | 3.89%               | Average error relative to each store's own sales level |
| R <sup>2</sup> | 0.9862              | Share of test-period variance explained                |
| Test records   | 1,170               | Final 26 weeks of each of the 45 stores                |
| Test window    | 4 May – 26 Oct 2012 | No overlap with the training period                    |